

It's hot

- The global home automation industry is projected to grow from \$16.9 billion in 2011 to \$35.6 billion by 2016, according to a study conducted by marketsandmarkets.com.
- International Data Corporation (IDC) anticipates there will be more than 200 billion connected devices by 2021, with more than 30 billion being autonomous devices.
- In a recent report published by Allied Market Research, it was concluded that the global smart home and buildings market is expected to grow at a compound annual growth rate (CAGR) of 29.5 per cent between 2012 and 2020.

What's inside?

- Sensors to know the environment
- Processor for intelligence
- Software for functionality, programmability and integration
- Connectivity for remote-accessibility, integration and diagnostics
- User interface on device or as a smartphone app for easy setup and usage

One example of a home energy management system is Savant's Smart-Energy monitoring solution, which sets your intelligent lighting to automatically turn on and off with the rising and setting sun by timing them to an astronomical clock, automates opening and closing of blinds to keep out the sun and conserve energy, regulates thermostat levels based on outside temperature, controls switching on or off of lights using motion sensors and lets you remotely manage your home's media components and other smart electronics using iOS devices to prevent prolonged operation and unnecessary energy consumption.

This, of course, is just the tip of the iceberg!

Always within reach

You might have thought your home became smarter the day the water heater supported auto cut-off or your kettle had temperature control or your microwave let you make *paneer butter masala* with its auto-cook option, but that is nothing compared to the possibilities that connectivity has opened up for smart devices!

Connectivity means many things. It means your devices can talk to each other, they talk to your mobile phone, they talk to (perhaps) a central server such as an energy monitoring system

at the utility end or a remote diagnostic solution of the original equipment manufacturer.

In short, the products will, over time, become a part of the Internet of Things, becoming intelligent and capable of managing their lifecycle by themselves to a large extent.

"M2M is at the core of the smart home offering. Effectively, it is one machine talking to another. It is the air-conditioner talking to the mobile phone. So, M2M is definitely a very critical part of the whole solution. From a protocol perspective, depending on what is being used, there are a bunch of protocols that are essential. For instance, all of the connectivity protocols such as 802.11, Bluetooth, Zigbee, 2G, 3G and 4G are all very essential," says Thilak Kumar, regional engineering manager, Wind River.

M2M makes many things possible. For example, as you walk along a passage, the lights could talk to each other

to ensure that subsequent lights turn on as the ones you cross go off.

Your bed could sense and tell the air-conditioner when you are feeling too warm or too cold, so the temperature can be adjusted accordingly.

Your dishwasher could tell your motor to turn on and fill the overhead water tank when the water flow in the pipeline reduces.

Above all this, connectivity also makes it possible to control multiple devices using a common interface, as it can be quite hassling to manage a dozen home automation tools individually!

One example of such an application is the Revolv Smart Home Automation solution, which supports multiple protocols including Wi-Fi, Z-Wave and Insteon and lets you access and control varied devices such as the Nest thermostat, Sonos speakers, Philips Hue lightbulbs and more, using the Revolv app on your smartphone.

That said, Kumar warns us, "Connectivity allows full control remotely by the owner, but at the same time brings in security risks that need to be addressed."

Home in your palms

"I believe that smart homes need to be looked at from the lens of mobility and smartphones today. The smartphone has quite quickly become the centre of our lives and I see tremendous potential for it to become the centre of our homes, and that is the single biggest trend in this space," says Annie Mathew, director, Alliances and Business Development, BlackBerry India. "What smartphones bring to the table are wireless protocols such as NFC and Miracast that are driving device connectivity. That said, they also bring a plethora of apps that create a singular console to manage home automation."

There are many popular mobile-based



Blacksumac's 'Piper' smart home automation and security device